

Bare-Metal

Level 1: Step 2
HARD
Theme: Virtualization and Operating Systems
Due Date: 18/10/2022

Please take the following notes in consideration:

- In this assignment the questions marked with a (*) are practical and have to be shown to the teacher.
- The remaining questions are to be answered in a written report to be delivered to the teacher.

Concepts

1. Consider the execution of type 1 virtualization platforms (bare-metal) nested in type 2 virtualization platforms (hosted).
 - (a) What possible interest could this approach have?
 - (b) What are the requirements for the execution of Hyper-V (type 1 platform) in VMware Workstation (type 2 platform), considering that Hyper-V has to support 64 bits VM?
2. Would you consider all the isolation modes provided by docker for windows, **OS-Level Virtualization solutions**? Explain why.
3. What are the possible sources of IP configuration for a VM's network adapter configured with a bridged connection

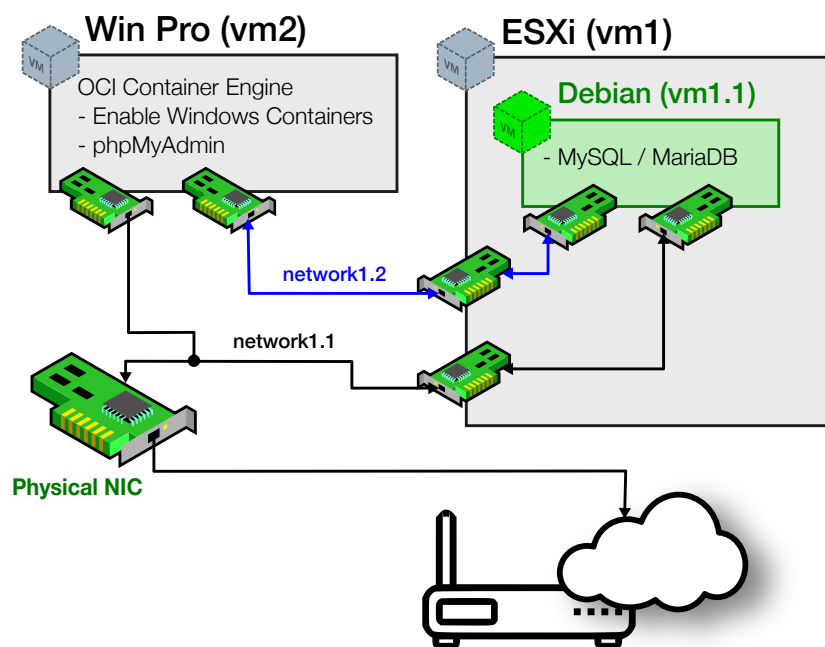
Virtualization

4. (*) **VMware ESXi**
 - (a) Create a **Virtual Machine** (vm1) and install **VMware ESXi** on it, respecting the following virtual hardware configuration:
 - 2×CPUs with, at least, 4GB of RAM and 1×16GB disk
 - 1 network interface (network1.1) - to provide **Internet Access** to the nested VMs;
 - 1 network interface (network1.2) **accessible only to your computer** - used to expose services running on the VMs.
 - (b) Investigate which IP address/network is in use for network1.2.
 - (c) Install a Debian **VM** (vm1.1) inside ESXi without a Desktop Environment and:
 - Ensure the nested VM has access to both network1.1 and network1.2.
 - Install and configure **MySQL** or **MariaDB** server on the VM:
 - Make sure the database server is accesible **only from network1.2**
 - Make sure the database root user can login from this same network

5. (*) **Docker**

- (a) Create a new **Windows 10 Pro**¹ VM (vm2) - **outside ESXi** - respecting the same virtual hardware parameters as above (4.a));
- (b) Install a **container engine** in the Windows VM you just created (5.a), and configure it to support either **windows containers** or **linux containers**;
- (c) Start a container with the official **phpMyAdmin docker image**;
- (d) Configure the **phpMyAdmin** docker image to point to the database server running on the vm1.1 created in 1.c)

6. Justify the container engine choice you made in 5.b).



◇ Keywords ◇

- os-level virtualization and containers
- open container initiative
- continuous integration
- windows container isolation
- virtual network modes

¹Use the official ISO image from <http://microsoft.com>

7. **Deliverables** (all deliverables submitted via [discord#assignments-submission](#)):

1. A PDF file with the assignment's report.
2. The file name should be 11s2h_aXXXXX_aYYYYY.pdf. This code means:
 - Level 1
 - Step 2
 - Hard difficulty
 - Students that participated in the group
 - **Maximum:** 2 students